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EEBus UC Implementation Guideline

Limitation of Power Consumption

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1 Scope of the document

This document is an implementation guideline for the Use Case "Limitation of Power Consumption" (short-name for this document: IG-LPC). It details how to interpret specific concepts and data points and gives examples and best practices on topics reported as probably difficult to understand or where misinterpretations could occur. It also puts the use case in the context of country-specific regulations.

The sections of this document clarify the [LPC] Use Case specification. They are, together with [LPC], the basis for testing procedures of EEBUS devices (e.g., in the Living Lab Cologne (LLC)).

Chapter 2 focuses on clarifications on the high-level part of the Use Case [LPC].

Chapter 3 focuses on clarifications on the technical part of the Use Case [LPC].

Chapter 4 defines country specific rules.

This document will be continuously updated as new issues requiring clarification are identified.

1.1 References

1.1.1 EEBUS documents

[SPINE]	*Minimum*: EEBus_SPINE_V1.3.0_Final.zip
[SHIP]	*Minimum*: EEBus_SHIP_TS_Specification_v1.0.1.pdf *Recommended*: EEBus_SHIP_TS_Specification_v1.1.0.pdf
[LPC]	EEBus_UC_TS_LimitationOfPowerConsumption_V1.0.0.pdf
[LPP]	EEBus_UC_TS_LimitationOfPowerProduction_V1.0.0.pdf
[UC-IG-General]	EEBus_UC_IG_GeneralGuidelines_V1.0.0.pdf
[SPINE-IG]	EEBus_SPINE_IG_ProtocolAndResourceGuidelines_V1.0.0.pdf

1.1.2 Normative references

[RFC2119]	IETF RFC 2119: 1997, Key words for use in RFCs to indicate requirement levels. Please see section 1.3.1 for details.
[BK6-22-300]	Addition to German law EnWG § 14a defining requirements for building's devices to be limitable by the DSO (27 November 2023).
[VDE-AR-E 2829-6]	The VDE-AR-E 2829-6 standardizes the communication interface for data exchange between customer systems and grid operators.

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1.2 Terms and definitions

ACK
Abbreviation for "Acknowledgement"

Actor
An Actor models a role within a Use Case definition (e.g. an energy manager or a heat pump).

BNetzA
Abbreviation for "Bundesnetzagentur (für Elektrizität, Gas, Telekommunikation, Post und Eisenbahnen)" (Federal Network Agency for Electricity, Gas, Telecommunications, Post and Railway), a German authority for all kinds of public grids and networks.

CEM
Abbreviation for "Customer Energy Manager". The CEM is an energy manager located at the user's home or premises or in a cloud application. The energy manager enables energy-optimised operation of the connected devices by harmonising energy demand and availability.

Control box
A control box can be part of a smart meter or of a smart meter gateway or a separate device. This depends on the country specific regulations, the local setup, etc.

Controllable System
Actor name for a logical or physical appliance or system that is remote controllable (e.g. via power limits).

CS
Abbreviation for "Controllable System"

EG
Abbreviation for "Energy Guard"

Energy Guard
Actor name for a logical or physical unit capable of managing the energy demand of connected devices.

EnWG
Abbreviation for the German law "Energiewirtschaftsgesetz" (Energy Industry Act)

EVSE
Abbreviation for "Electric Vehicle Supply Equipment"

DSO
Abbreviation for "Distribution System Operator"

EMS
Abbreviation for "Energy Management System"

GCP
Abbreviation for "Grid Connection Point"

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IG-LPC
Abbreviation for "Implementation Guideline for Use Case Limitation of Power Consumption" (short name of this document)

LLC
Abbreviation for "Living Lab Cologne" (see https://www.livinglabcologne.com for further information)

LPC
Abbreviation for the Use Case "Limitation of Power Consumption"

LPP
Abbreviation for the Use Case "Limitation of Power Production"

MGCP
Abbreviation for the Use Case "Monitoring of Grid Connection Point"

MPC
Abbreviation for the Use Case "Monitoring of Power Consumption"

NACK
Abbreviation for "No Acknowledgement"

Scenario
Part of a Use Case. Splitting a Use Case into Scenarios helps readers understand the Use Case more quickly. Some Scenarios are mandatory for a Use Case, whereas others may be recommended or optional.

SPINE
Abbreviation for "Smart Premises Interoperable Neutral-message Exchange": Technical Specification of EEBus Initiative e.V.

UC
Abbreviation for "Use Case"

1.3 Requirements

1.3.1 Requirements wording

The following keywords are used:

- SHALL
- SHALL NOT
- SHOULD
- SHOULD NOT
- MAY

Note: They apply only if written in capital letters.

For the meaning of the keywords, please refer to [RFC2119].

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1.3.2 Notes on requirements in this document

This document clarifies requirements of [LPC], [SPINE] (ProtocolSpecification and ResourceSpecification), or [SHIP]. Regarding these specifications, the keywords "SHALL" and "SHALL NOT" are already covered by mandatory requirements of these specifications.

Regarding other specifications (e.g. [BK6-22-300]), the keywords "SHALL" and "SHALL NOT" express mandatory requirements taken from these specifications or considered essential by the EEBUS Initiative e.V.

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2 Clarifications on the high-level part of the Use Case LPC

2.1 Intended purpose of the Use Case LPC

The Use Case [LPC] SHALL only be used for purposes specified in [LPC] (stabilization of the public electricity grid etc.; see section 2.1 in [LPC]). It SHALL NOT be used for other means like self-consumption optimisation, cost savings, etc. (for this, other Use Cases are available).

2.2 Duration of a limit

An Energy Guard SHALL NOT send an Active Power Consumption Limit with a duration set to "0".

The following rules clarify in detail situations with duration set to "0":

1) If Actor Controllable System receives an Active Power Consumption Limit with a duration set to "0", the limit SHALL immediately be deactivated by the Controllable System (see [LPC-007]). If the received write command contains a duration set to "0" AND also a request to set the Active Power Consumption Limit to "activated", the Actor Controllable System SHALL ignore the request to set it to "activated" and deactivate the limit instead*[2].
2) An Energy Guard is only permitted to send an Active Power Consumption Limit with a duration set to "0" if the limit is set to "deactivated" with the same command.
3) If an Energy Guard wants to deactivate an Active Power Consumption Limit, the limit can be deactivated explicitly with [LPC-008].

Note: If Actor Energy Guard does not want the Active Power Consumption Limit to depend on a duration value anymore, it needs to remove/unset the duration*[1]. Setting a duration to "0" does NOT remove/unset the duration!

*[1]: How the removal of a duration is done depends on the technical realisation of the Use Case. In case of SPINE, the according procedure is described in section 3.4.1.4 of [LPC].

*[2]: This means that the Controllable System SHALL NOT send a NACK to the Energy Guard because of the duration set to "0" together with the "activated" Active Power Consumption Limit (i.e., if no further issue is detected with the write command, an ACK is returned to the Energy Guard).

2.3 Heartbeat and timeout to detect missing Active Power Consumption Limit

The Use Case [LPC] specifies situations where Actor Controllable System receives a heartbeat but no following Active Power Consumption Limit (within a given duration). The following excerpt from [LPC] gives an example for such a situation:

> "*The Controllable System MAY leave the "failsafe state" and switch into "unlimited/autonomous" state if the Energy Guard's Heartbeat is received again, but no write command on the Active Power Consumption Limit is received within 120s ([LPC-921]).*"

(in section 2.3 of [LPC] this is enumerated as transition 10).

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This rule means that every heartbeat that is received by Actor Controllable System in "failsafe state" virtually instantiates its own timer to check if also an Active Power Consumption Limit is received latest 120s after the corresponding heartbeat was received or not.

This means that already the first heartbeat received in "failsafe state" permits Actor Controllable System to switch into state "unlimited/autonomous" if no Active Power Consumption Limit is received within 120s since this heartbeat.

It DOES NOT mean that a second heartbeat virtually disables the timer that belongs to the first heartbeat.

Note: The above-mentioned excerpt does not force Actor Controllable System to switch to state "unlimited/autonomous" if the condition is fulfilled. In this particular case, the Actor may well remain in "failsafe state" and may switch to state "unlimited/autonomous" later, unless another event occurs where rules of [LPC] require a specified action (e.g. the Actor Controllable System decides to remain in "failsafe state", but later a heartbeat and power limit within 120s are received wich require a switch to state "limited").

Of course, the explanations given above do not only apply to "transition 10", but to comparable situations as well.

2.4 ACK/NACK behaviour of an EMS as Controllable System

This section provides recommendations for an EMS as Actor Controllable System to report reliable ACK/NACK messages towards the grid's Actor Energy Guard.

The Use Case [LPC] clearly limits possibilities to reject (NACK) commands from an Energy Guard. However, it is also not valid for an EMS to accept (ACK) commands unconditionally. The following recommendations comply with [LPC] rules and provide a reliable ACK/NACK-based concept.

Note: A Controllable System sends an ACK message only if it will follow the limit. Otherwise, a NACK is sent. Hence, an ACK is NOT only an acknowledgement of receipt!

An example for the importance of the ACK concept is given in section 4.1.5.

As soon as an EMS receives a new activated power limitation, it has to evaluate whether the overall limit can be kept by the according devices*[1] or not. The best way an EMS can ensure this is by calcuating the limits for each device, sending the limits to the devices and analysing their responses (ACK = limit accepted or NACK = limit rejected). Then, the EMS can send a corresponding response to the grid's Actor Energy Guard (ACK, if the overall limit can be kept*[2] or NACK, if the overall limit cannot be kept).

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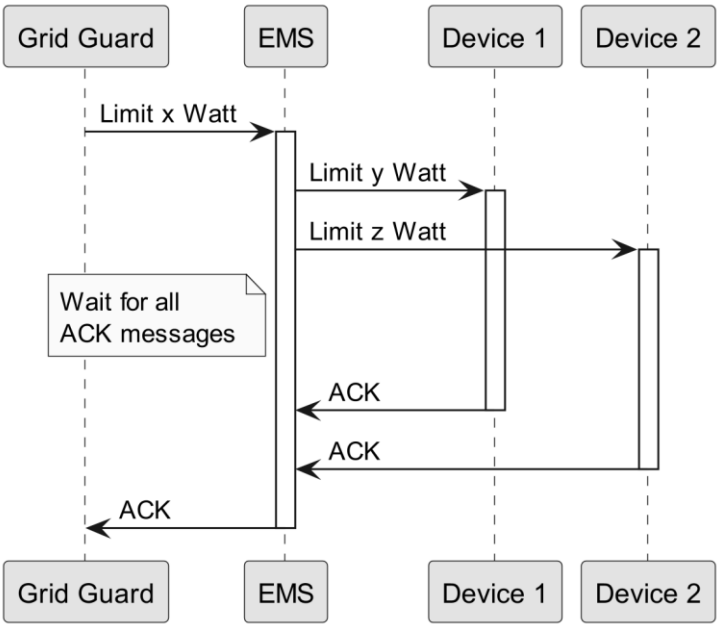


Figure 1: Example of EMS responding to grid's power limitation with full ACK-chain

However, there may be circumstances that prevent the EMS from receiving ACK/NACK messages from all relevant devices in time:

- A connected device's reaction time is too slow*[3]. Hence, the EMS cannot await the device's response until sending its response to the grid's Enery Guard as this must happen in a certain period of time*[4].
- Timing or (temporary) connection problems.

In such cases, the EMS needs to evaluate whether the overall limit can be kept or not by other means (e.g. by measuring the devices' power consumption, the power at the grid connection point, the devices' heartbeat messages, etc.).

Regardless of whether the EMS' response is based on ACK-messages from the devices or on the EMS' analysis of other available information, there might be the case that after the EMS acknowledged the grid's power limitation, the EMS later on determines that the overall limit cannot be kept. In such a case the EMS has to deactivate the grid's power limit and inform the grid's Actor Energy Guard by according means*[5] (see [LPC-923]).

*[1]: Which devices are affected may vary depending on the country specific regulations.

*[2]: Although one of the connected devices that is affected by the grid's power limitation has sent a NACK, the EMS possibly can ensure that the overall limit is kept by limiting another device even more.

*[3]: This problem may arise when a complex device, consisting of several components with one central unit that manages the components, their power consumption and the LPC-realisation, itself needs some time to evaluate whether the limit can be kept or not.

*[4]: In case of the SPINE protocol, the default value for the maximum reponse time is 10 seconds.

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*[5]: In case of SPINE-based implementations this is achieved with a subscription mechanism and proper notify-messages to the Energy Guard.

2.5 Energy Guard's reaction on NACK messages

The Use Case specification [LPC] does not described in detail how Actor Energy Guard needs to react on NACK messages received from Actor Controllable System as this is primarily the responsibility of the Energy Guard's business logic. However, this section will provide recommendations on specific aspects.

Keeping or removing a Controllable System: In case an Energy Guard receives a NACK message from the Controllable System, the Energy Guard SHOULD NOT remove the Controllable System from its list of devices that are used for its energy management. Instead, the Energy Guard should retry the rejected command (see below).

Retry after connection establishment: In [LPC] a case is described where a write command might be rejected due to timing circumstances. Excerpt from [LPC] section 2.2:

> "*Note: There may occur a timing problem when both CS and EG reboot almost the same time, but the CS finishes its reboot earlier: It may happen that the EG sends a Heartbeat and shortly thereafter a write on the Active Power Consumption Limit within 120s after the EG completed its reboot, but the CS - due to its earlier completion of the reboot - only recognizes the Heartbeat within its 120s period of state "init", switches then to state "unlimited/autonomous" (according to [LPC-906]), and receives the write afterwards. This results in a rejection of the write command just for timing reasons. Therefore, if the EG receives a reject on the write command within the assumed state "init", the EG may simply send Heartbeat and write command once more to make sure the rejection was caused by a timing problem and not because of a permitted exceptional reason.*"

In this case, the Energy Guard SHOULD retry its write command as described.

Retry or reschedule in other cases: Apart from situations described above, the Energy Guard may as well retry its write message to the same device, but wait a reasonable time for the first retry (e.g. 1 minute) and extend the time with every retry (e.g. one minute longer for every retry). However, it might also be that an Energy Guard needs to impose stronger power limitations to other devices to compensate the rejections of one device.

2.6 Actively control a Controllable System's Active Power Consumption Limit

As soon as Actor Energy Guard is configured to control an Actor Controllable System according to Use Case Limitation of Power Consumption*[1], the Energy Guard SHOULD do its best effort to keep Actor Controllable System in the states "limited" and "unlimited/controlled".

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*[1]: For a SPINE-based implementation this means that Actor Energy Guard acquired necessary binding permissions from Actor Controllable System.

Note: In theory, the Use Case Limitation of Power Consumption may work well without explicit communication with Actor Energy Guard as long as there is no real limitation of the power consumption required. However, as [LPC] is an essential Use Case to prevent critical grid conditions, it is necessary to monitor the communication chain in order to detect communication problems in time. Keeping an Actor Controllable System in the states "limited" and "unlimited/controlled" is the best approach to monitor the functional communication between these actors and to prove the functionality towards regulatory authorities.

2.7 Behaviour of EMS without grid limit provider

This section extends section 2.6 for the case that an EMS with Actor Energy Guard in "LPC instance 2"*[1] has not yet been configured to receive an (external) Active Power Consumption Limit from the grid (DSO). In terms of [LPC] this means that the EMS' role as Actor Controllable System in "LPC instance 1"*[1] does not know its responsible Actor Energy Guard so far. Therefore, the EMS cannot take over its responsibility to limit the building's power consumption according to [LPC]. In this situation, an EMS SHOULD behave as follows:

- If the EMS as Actor Energy Guard is already configured to control devices according to [LPC]:
  - The EMS SHOULD keep the devices in the state "unlimited/controlled".
- If the EMS as Actor Energy Guard is not yet configured to control devices according to [LPC]:
  - The EMS SHOULD NOT try to control yet uncontrolled devices according to [LPC]*[2,*3].

*[1]: For details on the "LPC instance 1/2", please refer to [LPC], section 2.4.

*[2]: For a SPINE-based implementation this means that the EMS SHOULD NOT acquire necessary binding permissions from potential Actor Controllable Systems.

*[3]: An uncontrolled device (Controllable System) will then switch to the state "unlimited/autonomous" 120s after the "init" phase, as there is no Energy Guard controlling it.

In theory, a building may contain several devices with EMS functionality including [LPC]. In this case, it is advised to first select the finally responsible EMS and configure it to receive limits from the grid. Afterwards, the EMS can then be configured to control devices.

2.8 Consideration of internal or remote communication problems of a Controllable System

This section does NOT address communication problems between Actor Energy Guard and Actor Controllable System. Rather, this section addresses the behaviour of a Controllable System for the following case:

- The communication interface between the Energy Guard and the Controllable System is active, but communication problems "within" the Controllable System prevent that commands received

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from the Energy Guard to limit the power consumption don't lead to a proper power consumption limitation at the power consuming components of the Controllable System.

Example 1: The communication interface of the Controllable System is realised by a hardware component "A" that implements the communication protocol with the Energy Guard. The power consuming component "B" of the Controllable System is a separate component with a separate communication to component "A".

Example 2: Like example 1, but component "A" or "B" requires an additional communication with the vendor's cloud service.

The case described above can be seen as a malfunction of the Controllable System regarding the Use Case [LPC]. Of course, other kinds of malfunctions may as well cause problems including a power consumption beyond the permitted limit. But the key point of the described case is that communication problems "within" a Controllable System are usually easily detectable by the Controllable System. This is esp. relevant as regulation authorities can impose rules on communication paths to

- ensure electricity grid stability even if communication problems with vendor's cloud services arise,
- ensure that users cannot easily bypass electricity grid stability measures by interrupt of easily accessible or configurable communication paths.

It is therefore recommended to detect "internal" communication problems and initiate safety mechanisms like the failsafe state of [LPC] in the present circumstances.

2.9 Failsafe Duration Minimum as shared data point

The data point Failsafe Duration Minimum of [LPC] is identical to the Failsafe Duration Minimum of [LPP]. This means both Use Cases SHALL always apply the same value.

If Actor Controllable System supports both Uses Cases [LPC] and [LPP], the Actor SHALL accept an Energy Guard's write command on the Failsafe Duration Minimum if any of the state machines of [LPC] or [LPP] currently permit writing the Failsafe Duration Minimum*[1], provided that the Failsafe Duration Minimum is within the permitted value range.

*[1]: The state machines of [LPC] and [LPP] are independent from each other. [LPC] and [LPP] specify a few conditions when a write on the Failsafe Duration Minimum shall not be evaluated. As long as one of the Use Cases permits writing the Failsafe Duration Minimum, it does not matter if the other Use Case does not evaluate the write command. This also means a write on the Failsafe Duration Minimum SHALL be rejected if neither [LPC] nor [LPP] permit the write command.

2.10 Recommendations on change (write) command frequency

Although [LPC] does not impose restrictions on the frequency of updates of the Active Power Consumption Limit (or other data points), it is strongly recommended that an Energy Guard SHOULD

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NOT write [LPC] data points more often than once every 5 minutes (except for retries due to errors or timing problems).

2.11 Order of messages after initial communication or communication loss

To ensure a safe state and prevent inconsistent configurations, a CS enforces a strict sequence during the initial communication phase with an EG. Any write command that deviates from this sequence SHALL be rejected by the CS.

The [LPC] defines the following rule in section 2.2:

> *In state "init" or "failsafe state" or state "unlimited/autonomous", only after a Heartbeat from the Energy Guard, a following received write command within 60 seconds on the Active Power Consumption Limit SHALL be evaluated. Earliest thereafter, write commands on any other data point defined in this Use Case SHALL be evaluated by the CS.*

Hence, a Controllable System will reject any write command on any data point defined within this Use Case (other than Active Power Consumption Limit) before a heartbeat is received followed by a write on the Active Power Consumption Limit. Implementations of the Energy Guard SHALL follow this order of messages.

Below, some more details are given.

Mandatory message sequence:

The following sequence SHALL be completed successfully before the CS enters a fully controllable state:

1. Establish heartbeat: The EG SHALL start sending its heartbeats.
2. Set initial limit: Following a heartbeat, the EG SHALL perform a write operation on the Active Power Consumption Limit within 60 seconds.
   Note: It is not strictly required that this write command follows the very first heartbeat, especially if other Use Cases demand higher heartbeat frequencies. The critical requirement is that a valid heartbeat must have been received within the 60-second window immediately preceding the write command).

Rejection rules:

Until the sequence above is finalized, the CS applies the following rejection logic:

- Pre-heartbeat phase: Any write command on any data point defined within this Use Case (including limits or configurations) is rejected.
- Post-heartbeat / pre-limit phase:
  - Write on Active Power Consumption Limit is accepted.
  - Write on any other data point defined within this Use Case (e.g., Failsafe Consumption Active Power Limit) is rejected.

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- Error handling: In case of a rejection due to an incorrect sequence, the CS SHALL respond with a NACK message.

2.12 Behaviour of an Energy Guard in case of missing heartbeats from the Controllable System

The functionality of the Energy Guard as pertaining to this Use Case SHALL NOT depend on the presence or absence of hearbeats from the Controllable System. The Energy Guard SHALL continue sending its own heartbeats as well as write commands on the Active Power Consumption Limit regardless of the state of a Controllable System's heartbeat functionality.

2.13 Behavior of an Energy Guard during connection establishment or restoration

When an Energy Guard (e.g., an EMS) establishes or restores a connection to a Controllable System, it SHALL ensure that grid stability is maintained at all times.

The following rules apply to prevent unintended violations of grid limits during the reconnection phase:

- An Energy Guard SHALL NOT send a "deactivated" Active Power Consumption Limit as the initial limit command following the restoration of communication if a power limitation is still required due to grid limitations.
- Sending a deactivation command immediately upon reconnection would allow the Controllable System to immediately leave its previous limited state or the failsafe state and increase its power consumption to its maximum nominal power.
- This behavior is prohibited because certain types of Controllable Systems (e.g., heat pumps) might be technically unable to immediately reduce their power consumption again after an allowed increase, even if the Energy Guard sends a corrected limit shortly thereafter.
- Therefore, the Energy Guard SHALL first send the currently required and activated Active Power Consumption Limit (either the previously active limit or an updated one) as the first write command following the heartbeat.
- A deactivation of the limit SHALL only be communicated if the current grid situation actually permits unlimited operation of the Controllable System.

2.14 Distinction between technical limit rejection and communication failures regarding failsafe state

To ensure grid stability, a clear distinction must be made between the rejection of a limit command due to technical constraints and a rejection caused by a violation of communication supervision rules (heartbeat).

- Communication integrity as prerequisite: Any procedure to leave the failsafe state (according to rules [LPC-916] to [LPC-920]) requires a verified communication link. This verification is only established if a valid heartbeat has been received within 60 seconds prior to the write command.

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- Scope of fallback rules: Rule [LPC-918] (transition to unlimited/controlled if a limit cannot be applied) is intended for scenarios where communication is functional and verified, but the specific limit data cannot be processed by the Controllable System (CS).
- Grid stability risk: If a CS receives a write command without a preceding heartbeat within the 60-second window while in failsafe state, the CS SHALL reject the command (NACK) without further evaluation of the limit's content.
- Reasoning: Without a recent heartbeat, the CS cannot ensure that the Energy Guard is operating correctly and remains in control of the local power constraints. Transitioning to an unlimited state while the control capability of the Energy Guard is unverified could pose a risk to grid stability at the grid connection point.
- Persistence of failsafe: In the event of such an "out-of-window" rejection, the CS SHALL remain in the failsafe state. It SHALL NOT leave this state until a valid sequence (heartbeat followed by a write command within 60s) has been successfully processed or the Failsafe Duration Minimum has been reached.

2.15 Write commands on failsafe values

The failsafe values "Failsafe Consumption Active Power Limit" [LPC-021] and "Failsafe Duration Minimum" [LPC-022] are critical parameters for ensuring grid stability in the event of a communication loss between the EG and the CS.

While the underlying technical data model provides the mechanisms for updating these parameters, the current high-level specification does not explicitly state a "SHALL" requirement for the CS to accept write commands on these specific data points. This lack of an explicit high-level mandate has led to misinterpretations where some implementations treat the changeability of failsafe values as an optional feature. This section clarifies that such support is mandatory for a compliant and safe implementation of the Use Case.

- Mandatory support for changeability: A Controllable System SHALL support the capability to receive and successfully process write commands on both failsafe data points. Although the [LPC] spefication states that the Energy Guard MAY change these values, this refers to the EG's discretion to initiate an update and does not allow the CS to reject such a command.
- Criticality of the failsafe power limit: The ability to adjust the "Failsafe Consumption Active Power Limit" is essential. If a CS were to prohibit changes to this value, the device would permanently fall back to the manufacturer's pre-configured default setting. Since these defaults might be set to high values (e.g., the maximum nominal power of the device), an unchangeable failsafe limit could fail to provide any meaningful grid protection. Such a restriction would undermine the objective of the failsafe mechanism and compromise the integrity of the entire Use Case. The EG must be able to configure this limit to match local grid constraints or contractual requirements.
- Failsafe Duration Minimum: Similarly, the "Failsafe Duration Minimum" SHALL be writeable by the EG. While the CS may enforce an upper limit for this value according to the specification (e.g., ensuring the duration is at most 2 hours), it must not generally block write access to this parameter.
- Alignment with testing: The requirement for the writeability of failsafe values is a mandatory part of device testing. The LPC high-level test specification includes explicit test cases for

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writing to these data points. A rejection of such write commands by a CS (unless the values are outside the permitted range) is considered non-compliant with the intended purpose of the Use Case.

2.16 Details on the Controllable System's ACK/NACK responses

Within [LPC], the following rules are defined regarding the Controllable System's ACK/NACK responses:

- Section 2.2:
  - If the limit can be applied, the CS SHALL inform the Energy Guard that the limit is accepted (ACK, [LPC-002/1]), otherwise the CS SHALL inform the Energy Guard that the limit is rejected (NACK, [LPC-003/1]).
  - Other messages (e.g., a write command on the Failsafe Duration Minimum) that are not accepted by the CS SHALL be declined with a NACK command ([LPC-003/2]).
- Section 2.6.1.1:
  - The Controllable System SHALL confirm an accepted limit (ACK, [LPC-002/1]). If the limit cannot be applied by the Controllable System, the Energy Guard SHALL be informed (NACK, [LPC-003/1]).
- Section 2.6.2.1:
  - The Controllable System SHALL confirm an accepted failsafe value (ACK, [LPC-002/2]). If the failsafe value cannot be applied by the Controllable System, the Energy Guard SHALL be informed (NACK, [LPC-003/2]).

Since there have been differing interpretations of when a value must be considered “accepted,” the following clarification is provided:

- A Controllable System SHALL accept (ACK) a write command on a data point if:
  - The complete write command is valid AND the Controllable System can apply*[1] the value.
- A Controllable System SHALL NOT accept (NACK) a write command on a data point if:
  - The write command contains an error (e.g., a negative value for the Active Power Consumption Limit) OR the Controllable System cannot apply*[2] the value.

*[1]: This means that the Controllable System is capable of following the Active Power Consumption Limit (APCL) or the failsafe value is within the permitted range. It does NOT mean that the value must be different from the already existing one or does affect the Controllable System at all.
Example 1: The current APCL is deactivated, the newly written APCL is deactivated as well: The CS can follow the deactivated limit -> Accept (ACK).
Example 2: The current APCL is activated (with a remaining duration), the newly written APCL is activated but with a duration of "0" (see section 2.2 as well): The CS can follow the limit with a duration of "0" (which results in an immediate deactivation of the limit) -> Accept (ACK).

*[2]: This means that the Controllable System is NOT capable of following the Active Power Consumption Limit (APCL) or the failsafe value is NOT within the permitted range.
Example 1: The newly written APCL has a value of "0", but the CS cannot follow this limit due to safety regulations (e.g., a heat pump being in defrost mode) -> Reject (NACK).

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Example 2: The newly written Failsafe Duration Minimum is greater than the CS's maximum value for the Failsafe Duration Minimum (see rule [LPC-022/4]) -> Reject (NACK).

2.17 Behaviour of the Controllable System's state machine in case of connection loss

The Use Case [LPC] supervises the communication between Actor Energy Guard and Actor Controllable System by means of heartbeats (see section 2.6.3 of [LPC]). Differing interpretations have been observed regarding the behaviour of the Controllable System's state machine (see section 2.3 of [LPC]) in case the communication connection between the Actors is interrupted. In particular, some implementations treated a connection loss as an internal teardown of the communication session that resets the Use Case state machine to state "init". This section clarifies that such behaviour is not compliant with [LPC].

Connection loss implies missing heartbeats: The transitions into the "failsafe state" are defined via the absence of received heartbeats. Excerpts from [LPC] section 2.2:

> *"The CS SHALL switch into "failsafe state" if it is in state "unlimited/controlled" and there is no Heartbeat received from the Energy Guard for at least 120 seconds since the last Heartbeat ([LPC-911])."*

> *"The CS SHALL switch into "failsafe state" if it is in state "limited" and there is no Heartbeat received from the Energy Guard for at least 120 seconds since the last Heartbeat ([LPC-912])."*

The conditions of [LPC-911] and [LPC-912] solely refer to the absence of received heartbeats. They do not depend on the state of the underlying communication connection and [LPC] does not define any exception for the case that the connection is lost. As no heartbeats can be received via an interrupted connection, the conditions of [LPC-911]/[LPC-912] are inevitably fulfilled 120 seconds after the last received heartbeat. This is intended, as section 2.1 of [LPC] explicitly names interrupted connectivity as a cause covered by the failsafe mechanism:

> *"If the Controllable System does not receive any Heartbeats ([LPC-031]) from the Energy Guard for more than 120 seconds (e.g. due to interrupted connectivity), the Failsafe Consumption Active Power Limit ([LPC-021]) is used as fallback. It is intended to prevent overloads in case of connectivity problems or when a building or device is repowered after loss of power."*

Hence, if the Controllable System is in state "limited" or "unlimited/controlled" and the communication connection is lost, the CS SHALL switch into the "failsafe state" according to [LPC-911]/[LPC-912] and SHALL remain in this state for at least the Failsafe Duration Minimum, unless rules [LPC-916] to [LPC-922] permit or require leaving the state. This applies regardless of the reason why heartbeats are missing and regardless of whether the connection was terminated properly (e.g. via the connection termination mechanisms of [SHIP]) or lost without notice (e.g. due to a network outage or a power loss of the remote device).

No transition to state "init" upon connection loss: The state machine of [LPC] enters state "init" solely after the completion of a (re)start of the Controllable System (see section 2.3.2 of [LPC]; enumerated as transition 0 in section 2.3.3 of [LPC]: "Restart of CS completed"). No transition from any other

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state to state "init" is defined. A loss, teardown or re-establishment of the communication connection is no "(re)start" of the Controllable System: Section 2.2 of [LPC] illustrates the meaning of a restart with the examples "switched off by the user" and "blown fuse", i.e. a restart of the system itself, not of its communication session or protocol stack*[1]. Consequently, upon a connection loss, the Controllable System SHALL NOT reset its state machine to state "init" and SHALL NOT treat an internal reset of its communication context as a "(re)start" in the sense of [LPC] section 2.3.2.

Reasoning (grid stability risk): In state "init", the Controllable System may switch into state "unlimited/autonomous" if no heartbeat and following write command on the Active Power Consumption Limit are received within 120 seconds ([LPC-906]). As no heartbeat can be received while the connection is lost, a Controllable System that (incorrectly) entered state "init" upon a connection loss would become unlimited approximately 120 seconds after any communication failure. An active power limitation would effectively end after ~120 seconds instead of being replaced by the failsafe limitation for at least the Failsafe Duration Minimum (2 to 24 hours). This would undermine the entire failsafe concept: Table 1 in section 2.2 of [LPC] requires the Controllable System to be "Limited by Failsafe Consumption Active Power Limit" as long as heartbeats are not received in time ([LPC-914/2]) and the Failsafe Duration Minimum has not expired. Furthermore, the failsafe state is intended exactly for situations where the Energy Guard is completely unavailable (cf. "broken Energy Guard" in section 2.6.2.1 of [LPC]) - an unavailable Energy Guard does not maintain a communication connection either.

Behaviour after restoration of communication: After the connection is re-established, the Pre-Scenario communication is performed as required (see section 3.3 of [LPC]), while the Controllable System remains in its current state of the state machine (typically the "failsafe state", if the interruption lasted longer than 120 seconds). The Controllable System then leaves the "failsafe state" according to the regular rules, i.e. upon receipt of a heartbeat and a following write command on the Active Power Consumption Limit (see [LPC-913] and [LPC-916] to [LPC-920]). A Controllable System is NOT required to be in state "init" when a connection is established: An Energy Guard can establish a connection to a Controllable System being in any state of the state machine.

Note: Deliberately restarting the Controllable System (or parts of it, e.g. its Use Case implementation) upon or after a connection loss in order to reach state "init" would circumvent [LPC-911]/[LPC-912] and is considered non-compliant with the intended purpose of the Use Case.

Alignment with testing: The LPC test specifications verify the transitions into the "failsafe state" with mandatory test cases that simulate an interrupted connection (e.g. by disconnecting the network). A Controllable System that resets its state machine to state "init" upon a connection loss will fail these test cases.

*[1]: For SPINE-based implementations: Discarding session-related communication artefacts (detailed discovery information, subscriptions, bindings) upon a connection loss and re-establishing them after reconnection according to section 3.3 of [LPC] is a common and compliant implementation approach. However, these communication artefacts are independent of the operational state of the Use Case state machine, which continues to apply across the interruption.

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3 Clarifications on the technical part of the Use Case LPC

3.1 Reaction to write commands outside the valid ranges

In case the Controllable System receives a write command from the Energy Guard that is out of the permitted value range*[1], the Controllable System SHALL send a NACK response. This means it SHALL NOT discard the message without informing the Energy Guard.

*[1]: E.g., a write on the Failsafe Duration Minimum ([LPC-022]) with a value higher than 24 hours as defined in [LPC-022/4].

3.2 Usage of Actors and entityTypes

For both Actors (Energy Guard and Controllable System) there is a list of Entity Types that are permitted (see sections 3.2.1.1.1 and 3.2.2.1.1 of the [LPC] UC).

For an EMS that implements both roles of the UC, the Entity Type "CEM" SHALL be used for the Actor Energy Guard and for the Actor Controllable System.

A control box only implements the Actor Energy Guard towards an EMS or end device. For this implementation, the Entity Type "GridGuard" SHOULD be used in order to differentiate it from an EMS in the same network.

3.3 Usage of featureType "Generic" for client functionality

Each Actor SHOULD use a client Feature with featureType "Generic" for all its client functionality of its respective Entity.

Regarding [LPC], an Actor SHALL implement its client and server Features in the same Entity.

Rationale: The techincal SPINE solution in [LPC] describes an Actor's server Features (Function tables)*[1]. With the Use Case discovery information of Actor A, the Entity of its server Features can be detected by the other Actor B*[2]. But it also means that the Actor's client functionality belongs to this Entity. Having client and server functionality in the same Entity is also essential for Features that require binding. The latter enables a server Feature to verify if a received write command contains a source address of a granted binding partner.

[1]: In some Use Cases a given Actor has only client functionality or only server functionality. Use Case discovery returns the Actor's Entity[2] regardless of its server or client functionality.

*[2]: Without going into detail, there are also Use Cases with a dynamic Entity where Use Case discovery can only report a fixed parent Entity. Nevertheless, the occurrence of a well-defined child Entity and its resources determines then the presence of the respective Actor.

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3.4 Usage of Element measurementId in loadControlLimitDescriptionData

Within the loadControlLimitDescriptionData (as part of the data point Active Power Consumption Limit, [LPC-011]), the Element measurementId is defined. This Element is marked as mandatory in the first column, but in the last column ("Element and value rules") the following is specified:

> "*SHALL be set as FOREIGN IDENTIFIER, if a measurand or other feature is linked with the measurementId. [...]*"

Despite the description, the "M" in the first column has a higher priority, i.e. the Element is mandatory in any case (not only if the Measurement Feature is available on the same Entity that supports an according measurand). Hence, an Energy Guard SHALL consider the content of loadControlLimitDescriptionData as invalid if the Element measurementId is not set. Hence, if a Controllable System does not provide a corresponding measurand, the Element measurementId SHALL still be set (e.g., to a (relatively high) random number that is not used by any measurand on this Entity).

3.5 Binding partner clarifications

The [LPC] UC specifies in section 3.4.1.1: "Only one binding partner is allowed to write the data specified in this Scenario.". This does not mean that only one binding partner is permitted in total for the device. It just means that the Controllable System SHALL NOT permit more than one of its binding partners to modify those data points that are changeable by the Energy Guard according to [LPC].

Example: We assume the Controllable System with Use Case [LPC] is just one of several Use Cases implemented on a SPINE device called "C". An Energy Guard on device "A" acquires binding permission on the "LoadControl" Feature of device "C". For another Use Case "UCX", device "B" needs as well binding permission on the same "LoadControl" Feature. This means the Feature contains entries for both Use Cases. The entries are distinguishable by their identifiers. In this case, device "C" needs to grant binding permissions to the "LoadControl" Feature for device "A" as well as for device "B". However, it must ensure that device "A" does not alter entries belonging to Use Case "UCX" which is controlled by device "B". Likewise, device "C" needs to prevent that device "B" alters data belonging to Use Case [LPC].

How the Controllable System determines which device to allow changing the [LPC] entries is implementation specific (e.g., the first device that attempts to write an [LPC] entry is permitted, any following device trying to write on the same entry is rejected).

Update with version 1.1.0 of this implementation guide:

With the upcoming SPINE V1.4.0, at maximum one binding partner will be permitted per Feature. This means that if one communication partner (device "B") requested a binding on a Feature on device "C" and that binding was accepted by device "C", no further binding request on this Feature will be permitted by "C".

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Due to race conditions, there might be a situation (e.g. after a power loss of the building) where device "A" and device "B" try to bind to device "C"'s Features simultaneously, but in a different order (e.g., device "A" first tries to bind to Feature "LoadControl" and device "B" first tries to bind to Feature "DeviceConfiguration"). Device "C" could now permit both binding requests, as they are each the first on the according Feature. But as the UC [LPC] relies on both Features, there would be a situation where none of the two devices "A" or "B" can operate the UC [LPC] correctly.

Therefore, the following rule SHOULD be applied:
If a device "A" successfully requested a binding on one of the Features of device "C" used by [LPC], device "C" SHOULD NOT permit any other device "B" a binding on any of the Features of the UC [LPC] that require a binding (only device "A" is permitted to bind on these Features then).

Note: At the time of release of version 1.1.0 of this implementation guide, the rule above is not subject to EEBUS certification (the rule described in the preceding section remains the basis for certification). With the adoption of SPINE V1.4.0 as the normative basis for [LPC], the SHOULD and SHOULD NOT requirements stated above will be replaced by SHALL and SHALL NOT requirements respectively. I.e., the rule will become strictly mandatory and testing-relevant. Implementers are therefore strongly encouraged to already apply the rule, to ensure forward-compatibility with SPINE V1.4.0.

Note: Please also consider section "Clarification on communication modes and mandatory device addresses" in [SPINE-IG].

3.6 Clarification on permitted value ranges for LPC data points

Section 2.8.1 of [LPC] defines that the data points

- Failsafe Consumption Active Power Limit ([LPC-021])
- Power Consumption Nominal Max ([LPC-041])
- Contractual Consumption Nominal Max ([LPC-042])

are always greather than or equal to zero. Even though these value ranges are not explicitly repeated within the technical part of [LPC], an EG SHALL only write values ≥0W for the listed data points. A CS SHALL react with a NACK to write commands on [LPC-021], [LPC-041] or [LPC-042] with values <0W.

3.7 Heartbeat Elements - presence indication and verification obligations

The heartbeat message contains the following Elements:

- timestamp
- heartbeatCounter
- heartbeatTimeout

As a server instance, all three Elements SHALL be set to values as specified in [LPC].

A client instance, on the other hand, is not obliged to verify the presence or the correct content of the Elements heartbeatCounter and heartbeatTimeout.

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The above-mentioned rules guarantee a correct content of the heartbeat messages, but reduce the workload of a client implementation.

3.8 Behaviour of Controllable System in case of multiple Energy Guard instances on a connected device

Some implementations of Energy Management Systems (EMS) expose multiple instances of the Actor Energy Guard (Entity Type "CEM") on different Entities within the same device due to technical reasons. During the discovery phase, it is often not apparent to a Controllable System which of these instances is the active control instance for the [LPC] Use Case.

To ensure deterministic behaviour in this scenario, the following rules apply to the Controllable System:

Precondition: These rules only apply if the Controllable System detects more than one Entity with the Actor "CEM" supporting the [LPC] Use Case on a connected device.

Procedure:

1. Delay of heartbeat subscription: The Controllable System SHALL NOT send a subscription request to the DeviceDiagnosis Feature of any of the discovered CEM Entities immediately after discovery, as it is unclear at this stage which Entity assumes control.

2. Await binding requests: The Controllable System SHALL wait until it receives binding requests from the Energy Guard. The Energy Guard must request bindings for the Features LoadControl (for power limits) and DeviceConfiguration (for failsafe values) to obtain write privileges.

3. Validation of Entities: The Controllable System SHALL verify whether the binding requests for the Features LoadControl and DeviceConfiguration originate from the same Entity (i.e., the same source address).
    - If the requests originate from the same Entity, this Entity is considered the valid Energy Guard instance.
    - If the requests originate from different Entities, the Controllable System SHALL consider this state as an error and SHALL NOT accept the bindings (or SHALL terminate further communication for the [LPC] Use Case).

4. Execution of subscription: Only after the bindings for LoadControl and DeviceConfiguration have been successfully established with the same Entity, the Controllable System SHALL send the subscription request for the DeviceDiagnosis Feature (Heartbeat) to exactly that Entity.

This procedure ensures that heartbeat monitoring and active control (limits) are consistently handled via the same logical instance (Entity).

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3.9 Multiple LPC instances on one device

This section needs to be separated into paragraphs for multiple [LPC] instances as a Server Actor (Actor Controllable System) or a Client Actor (Actor Energy Guard).

Note: For a definition of "Server Actor" and "Client Actor", please refer to [UC-IG-General].

In general, it is not recommended to have multiple [LPC] instances of the same Actor on one device.

Multiple [LPC] Server Actor instances with Actor Controllable System:

For a Client Actor, it must always be unambiguous which [LPC] Server Actor instance it needs to control. This can be achieved by the following rules:

- On the highest Entity hierarchy level, there SHALL be at most one [LPC] Server Actor instance.
- On nested Entity hierarchy levels, there MAY be several [LPC] Server Actor instances, but only if there is a parent Entity level with at most one [LPC] Server Actor instance.

Device hierarchy example:

-- device "A"
---- entity "1"
------ entity "1.1"
------ entity "1.2"
---- entity "2"

Permitted [LPC] Server Actor instances for the given device hierarchy example:

1. [LPC] Server Actor on entity "1" and on entity "1.1" and "1.2"

Not permitted [LPC] Server Actor instances for the given device hierarchy example::

1. [LPC] Server Actor on entity "1" and on entity "2"
2. [LPC] Server Actor on entity "1.1" and on entity "1.2"
3. [LPC] Server Actor on entity "2" and on entity "1.1" and "1.2"

Multiple [LPC] Client Actor instances with Actor Enery Guard:

For a Server Actor, it must always be unambiguous by which [LPC] Client Actor instance it will be controlled. This can be achieved by the rules stated in section 3.8.

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3.10 Extension of the permitted Entity Types

With version 1.1.0 of this Implementation Guide, the following Entity Type is added to the list of permitted Entity Types for the Actor Controllable System:

- BatterySystem

An Actor Energy Guard SHOULD accept this Entity Type as valid for the Actor Controllable System of a communication partner.

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4 Country specific rules for the Use Case LPC

4.1 German EnWG § 14a (2024)

With the release of [BK6-22-300] as part of the EnWG § 14a, the BNetzA (*Bundesnetzagentur für Elektrizität, Gas, Telekommunikation, Post und Eisenbahnen*; Federal Network Agency for Electricity, Gas, Telecommunications, Post and Railway) defined requirements for specific devices in the buildings that need to be limitable by the DSO. The following kind of devices are listed:

- non-public charging stations
- heat pumps (including backup heaters)
- air conditioners
- stationary batteries (inverters).

These devices are called "SteuVEs" (*Steuerbare Verbrauchseinrichtungen*; controllable appliances)

The following sections detail how the Use Case [LPC] can be used to fulfil the new § 14a requirements.

For further details please refer to [BK6-22-300].

Please note that new versions of the law and its additional documents may be released that may change the regulations that are quoted within this document.

4.1.1 Interpretation of LPC limit in Germany

As with the [BK6-22-300] only specific kinds of devices are obliged to follow the DSO's limits, an [LPC] Active Power Consumption Limit does not limit the building's complete consumption at the grid connection point (GCP), but only the consumption of the "SteuVEs".

If more than one "SteuVE" is installed in the building and they are managed by an EMS (this means, they are not individually connected to a control box), the limit that is sent to the EMS limits the combined power consumption of all installed "SteuVEs" (see section 4.1.2, option 2, for details).

4.1.2 Direct vs. EMS controlled limitation

There are two possibilities how devices are limited via [LPC]:

1) Directly via the control box
2) Via an EMS that receives a limit from the control box and sends proper limits to the end devices

With option 1, the device is connected to the control box and the DSO addresses the device explicitly. If further devices are connected, each receives its own limit. With this option, a building's PV system cannot be used to enable a higher power consumption of the device as it is not taken into account.

With option 2, an EMS receives a limit from the control box. The EMS has to ensure that all SteuVEs together do not consume more power from the grid than permitted by this limit. A local PV system

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can be used to permit higher power consumption of the end devices, as the self-produced power reduces the load at the grid connection point. Additionally, the EMS may allocate the power of the different devices individually (based on the customer's needs), as long as the overall limit is kept.

4.1.3 Minimum values for limits

In [BK6-22-300] it is specified that devices are permitted to use a certain amount of power even if a power limitation is active.

It has to be differentiated between the different kind of devices and the amount of devices in the building. For non-public charging stations and stationary batteries different calculations are specified as for heat pumps and air conditioners.

Details can be retrieved from the official appendix no. 1 of the document [BK6-22-300].

4.1.4 SPINE entityTypes for the SteuVEs

The following table lists the SPINE entityTypes that SHALL be used by Actor Controllable System on a SteuVE:

SteuVE device category	SPINE entityType
non-public charging stations	EVSE
heat pumps	HeatPumpAppliance
air conditioners	HeatPumpAppliance*[1]
stationary batteries	Inverter

Table 1: SPINE Actors for the SteuVEs

*[1]: As air conditioners are not part of the [LPC] specification yet, the Actor HeatPumpAppliance SHALL be used instead (as most air conditioners are technically similar to heat pumps).

Additionally, an EMS can be used to receive the DSO's limitations. In this case, the entityType "CEM" SHALL be used by the EMS.

4.1.5 SPINE ACK/NACK messages as provision of proof

The EnWG § 14a requires that the device operator (end customer) can prove the fulfillment of a power limitation. Apart from permitted exceptions and transition rules, this requirement affects each installation, regardless of whether the affected devices and their applications and protocol stacks can already contribute to this proof or not. In case of implementations based upon SHIP/SPINE with the SPINE-based realisation of [LPC], the BNetzA currently considers the requirements for information exchange and verification of grid-oriented control to be already fulfilled (by refering to the [VDE-AR-E 2829-6]).

Background information: Although the high-level part of [LPC] already describes a Controllable System's acceptance (ACK) or rejection (NACK) of Energy Guard commands, these mechanisms are described in a technology independent way, hence can be realised in various ways. To support a robust signalling of acceptance and rejection of commands, SPINE provides "result" messages to transport "ACK" and "NACK" responses. Additionally, commands and responses are linked together

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with message numbers in SPINE. In combination with a transport protocol like SHIP, this leads to a reliable response mechanism for the Energy Guard's commands. This mechanism, together with the SPINE realisation of [LPC], fulfils the above-mentioned requirements.

As described in section 2.4, an EMS needs to evaluate whether to send an ACK or NACK back to the grid's Actor Energy Guard. In case the EMS received responses (ACK/NACK) from the affected devices and based its response on them ("full ACK-chain"), the EMS does not need to provide additional logging for the provision of proof to the BNetzA. In contrast, if the EMS' response is based on other information (e.g., the devices' power consumption, the power at the grid connection point, etc.), the EMS must be able to provide this information in case it needs to prove its decision (i.e., it must log this information).

The ACK/NACK messages received at the DSO's Energy Guard (e.g. a control box) via the SPINE protocol are reported as appropriate information to the DSO's backend where the success or failure of the write attempts can be logged for later evaluation and provision of proof to the BNetzA.

4.1.6 Maximum reaction time for power limitation

The [BK6-22-300] distinguishes between the reaction time of a DSO and the reaction time of a device operator (end customer). Broadly speaking, a DSO must send a power limitation to each affected device operator not later than 5 minutes after the detection of the necessity to limit the power consumption of the respective part of the electricity grid. For the device operator, the [BK6-22-300] requires an "immediate" fulfillment of the received power limitation. However, the requirement to react "immediatly" is mitigated by typical reaction times of a system/devices (i.e. times that can be expounded) as well as necessary delays that would otherwise damage the system/devices. For stationary batteries (inverters) and non-public charging stations, the reaction time to be expected is within seconds, whereas a reaction time of a few minutes can be required by electromechanical devices like heat pumps or air conditioners.

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